

Adama KOMI



Software Engineer - Fullstack & Mobile Developer

☎ +212 619 124 131 ✉ adamakomi15@gmail.com 📍 Mohammedia, Morocco

LinkedIn : linkedin.com/in/adama-komi GitHub : github.com/adamaKomi

ABOUT

Graduate in Software Engineering (FST Mohammedia), I am a **Software Engineer and Fullstack & Mobile Developer** passionate about designing, developing, and deploying robust, scalable, and high-performance web and mobile architectures. With hands-on experience gained through internships and ambitious projects, I am proficient in building business and consumer applications using **Node.js (React, React Native, NestJS)**, **Python (FastAPI, Qdrant/AI)**, and **Java (Spring Boot)** ecosystems. I am seeking a permanent position and am immediately available.

TECHNICAL SKILLS

- **Mobile Development:** React Native, Expo, Android & iOS
- **Backend Development:** Node.js (NestJS, Express), Python (FastAPI, Flask), Java (Spring Boot), PHP, C/C++
- **Frontend Development:** React (v19), TypeScript, JavaScript, Tailwind CSS (v4), HTML5/CSS3, Material-UI, Bootstrap
- **Databases & ORM:** PostgreSQL, Prisma ORM, TypeORM, SQLAlchemy, MySQL, MongoDB, Qdrant (Vector DB)
- **DevOps & Tools:** WebSockets (Socket.io), Redux Toolkit, BullMQ, Celery, Redis, Docker, Docker Compose, Git/GitHub, CI/CD (GitHub Actions), Firebase Admin SDK, WordPress, Postman
- **Architecture & Practices:** Clean Architecture, Domain-Driven Design (DDD), Hexagonal Architecture, REST API, Tests (Pytest, Jest, Vitest), Agile Methodologies (Scrum)

PROFESSIONAL EXPERIENCE

✓ Editorial SaaS Platform

Final Year Internship (Thesis) | Atrait Technologies

March 2026 – Present

Development of an editorial automation SaaS platform enabling researchers to automatically analyze and format manuscripts (.docx) and images according to scientific journal guidelines.

- **Backend & Asynchronous Processing:** Design of the backend architecture and REST API with FastAPI, and implementation of asynchronous background tasks (file ingestion, parsing, validation) using Celery and Redis.
- **AI Recommendation Engine (NLP):** Implementation of a hybrid recommendation algorithm leveraging semantic search with the Qdrant vector database and PubMedBERT embeddings.
- **Image Processing Pipeline:** Automated quality control of figures (DPI validation, JPG/PNG to TIFF/EPS conversion) using Pillow.
- **Dynamic Frontend:** Design of a step-by-step workflow UI with React 19, Vite 8, TypeScript, and TanStack Query.

✓ Web Developer

End-of-Year Internship (PFA) | Atrait Technologies

April 2025 – July 2025

Contribution to the development and customization of an e-commerce platform based on the WordPress CMS.

- **PHP Development & Extensions:** Design and development of custom features and plugins with PHP to adapt the site behavior to specific business needs.

- **Integration & Themes:** Advanced theme customization, creation of dynamic and responsive web pages with HTML5, CSS3, and JavaScript.
- **Data Management:** Administration and querying of a MySQL database to ensure data integrity and persistence (orders, product catalog, user accounts).

NOTABLE ACADEMIC PROJECTS

✓ Parkour Mobile App – Tracking & Gamification

FST Mohammédia

November 2025 – January 2026

Full design and development of a cross-platform mobile application for physical activity tracking and gamification.

- **GPS Tracking & Sensors:** Implementation of background geolocation tracking via Expo Location / Task Manager, live route tracing with MapLibre React Native, and step counting using the accelerometer (Expo Sensors).
- **WebSocket & Real-Time:** Development of a real-time communication module to transmit running coordinates, and integration of an anti-drift validation algorithm on the backend.
- **Backend & Data Architecture:** Creation of a modular server with NestJS connected to a PostgreSQL database via TypeORM, and push notification management with Firebase Admin SDK.
- **Gamification & State:** Application state management with Redux Toolkit and implementation of interactive features (dynamic podium, weekly leaderboard, on-the-fly badge system).

✓ QueueClock Web & Mobile App – Virtual Queue Management

FST Mohammédia

September 2024 – January 2025

Design and development of a distributed monorepo virtual queue management system in real time, comprising a client mobile app, web administration portals, and a modular API.

- **Backend & Concurrency:** Backend development with NestJS connected to PostgreSQL via Prisma ORM. Implementation of transactional locks and raw SQL index constraints to secure counter sessions.
- **WebSockets & Real-Time:** Integration of Socket.io coupled with a Redis adapter to instantly propagate queue positions and wait time estimates to clients and agency displays.
- **Asynchronous Tasks:** Use of BullMQ and Redis for automatic asynchronous detection and invalidation of geographically conflicting tickets.
- **Client & Mobile Applications:** Development of a hybrid mobile app in React Native (Expo) and Redux Toolkit with network resilience, and web administration portals in React and TypeScript.

EDUCATION

State Degree in Software Engineering and Integration of Computer Systems

June 2026

Faculty of Sciences and Technologies (FST) of Mohammedia, Morocco

University Degree in Sciences and Technologies (DEUST) – With Honours

June 2023

Faculty of Sciences and Technologies of Béni Mellal, Morocco

Scientific Baccalaureate – With Highest Honours

June 2021

Lycée Municipal de Titao, Burkina Faso